Customer Lifetime Value (CLV/LTV) Prediction Model

Project Report

Abstract

Customer Lifetime Value (CLV) is a key business metric that estimates the total value a customer is expected to generate over the course of their relationship with a company. This project builds a machine learning model that predicts CLV using an e-commerce transaction dataset (Online Retail II). Raw transaction records were cleaned and aggregated to the customer level, and behavioral features — Recency, Frequency, Monetary Value, Average Order Value (AOV), Total Quantity Purchased, and Purchase Span — were engineered as predictors. Random Forest and XGBoost regressors were trained and compared, and the better-performing model was used to predict LTV for every customer and segment them into Low, Medium, and High value groups to support targeted marketing and retention strategies.

Introduction

Customer retention is critical to sustainable business growth, and acquiring new customers is typically more expensive than retaining existing ones. However, many organizations struggle to identify which customers are likely to generate the most long-term value, making it hard to allocate marketing budgets efficiently. This project addresses that gap by building a CLV prediction model from historical online retail transaction data, grouping purchases by Customer ID and engineering features that capture how recently, how often, and how much each customer spends. The resulting predictions allow the business to identify its most valuable customers and design differentiated engagement strategies for each customer segment.

Tools Used

Category	Tools / Libraries
Language & Environment	Python, Jupyter/Colab Notebook
Data Handling	pandas, numpy
Visualization	matplotlib, seaborn
Machine Learning	scikit-learn (Random Forest Regressor, train-test split, metrics), XGBoost (XGBRegressor)
Model Persistence	joblib

Steps Involved in Building the Project

- **Data Understanding:** Loaded the Online Retail II transaction dataset and reviewed its shape, structure, data types, duplicate rows, and missing values (mainly in Customer ID and Description).
- **Data Cleaning:** Removed rows with missing Customer IDs and Descriptions, dropped duplicate records, excluded cancelled invoices (Invoice numbers starting with 'C'), and filtered out invalid (non-positive) Quantity and Price values. Converted InvoiceDate to datetime and created a Sales column (Quantity × Price).
- **Feature Engineering:** Aggregated cleaned transactions by Customer ID to build a customer-level dataset, then derived Recency (days since last purchase), Frequency (number of unique invoices), Monetary value (total spend), Purchase Span (days between first and last purchase), and Average Order Value (AOV).
- **Model Building:** Split the customer-level data into training and test sets (80/20) using Recency, Frequency, AOV, Purchase Span, and Total Quantity as features and Monetary value as the target (LTV proxy). Trained a Random Forest Regressor and an XGBoost Regressor.
- **Model Evaluation:** Compared both models using MAE, RMSE, and R^2 Score to select the better-performing model (XGBoost) for final predictions.

- **Prediction & Segmentation:** Used the trained XGBoost model to predict LTV for all customers, then segmented customers into Low, Medium, and High Value groups based on predicted LTV quantiles.
- **Model Saving & Export:** Saved the trained model with joblib and exported final predictions to ltv_predictions.csv. Analyzed feature importance to identify which behaviors most influence predicted LTV.

Conclusion

This project successfully developed a Customer Lifetime Value prediction model using historical online retail transaction data. After cleaning the data and engineering customer-level behavioral features, Random Forest and XGBoost regressors were trained and evaluated, with XGBoost selected for final predictions. Customers were segmented into Low, Medium, and High value groups, showing that Total Quantity, Recency, and Frequency are the strongest drivers of predicted LTV. These insights enable the business to prioritize high-value customers for retention and loyalty programs, nurture medium-value customers through upselling, and design cost-effective re-engagement campaigns for low-value customers — supporting more efficient allocation of marketing resources.